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Network Clock Frequency Synchronization

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Abstract

The emergence of network convergence emphasizes the need to support distributed synchronous servers such as TDMoIP (pseudo-wire) and 3G cellular gateways over a packet switched infrastructure. Consequently, we formalize the problem of network wide clock frequency synchronization and introduce novel and efficient algorithms to synchronize the frequency among all the clocks in the network with respect to a single frequency.

The common thread of our solutions is that they take a network-wide view that accounts for all the clocks in the network and measurements taken over all links to estimate the frequency difference of each clock with respect to the reference clock. The various presented algorithms introduce different trade-offs between the accuracy and the computation complexity. While all our schemes are global, they employ simple pair-wise measurements between neighboring nodes. Consequently, all the algorithms presented in the paper, are simple, easy to implement and require a modest amount of measurement and control traffic.